



UNIVERSITY OF AMSTERDAM

Tutorial 2 / Part 2 :

Natural Language Processing & Text Analytics:

Sentiment and Topic Analysis with Evaluative Lexicon, Textblob & Hugging-Face (Zero-Shot Classifier)

Digital Marketing and Analytics: Week 1 Tutorial 2 (Part 2)

Academic Year 2025-2026 – Semester 1 Block 2

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Unstructured Text Data: Sentiment & Topic Analysis

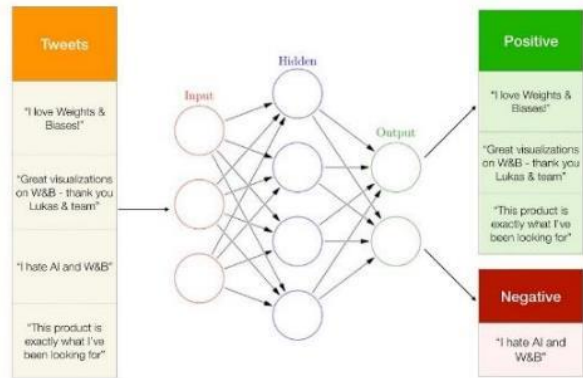
Tekst Analytics is a broad domain in data science and many methods-approaches available.

In Digital Marketing practice we mainly use two methods – with the help of Natural-Language-Processing / Machine Learning.

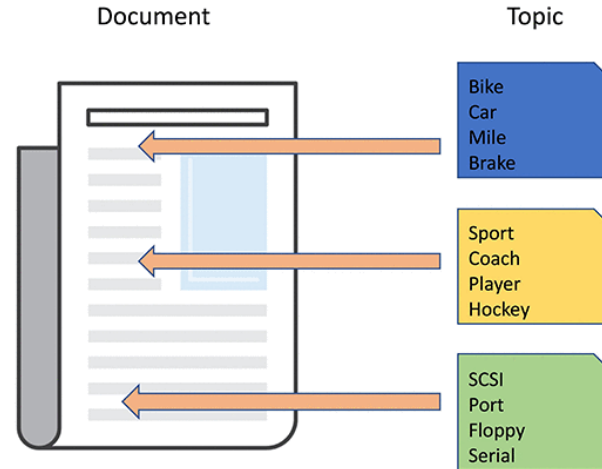
1 Sentiment (Emotion-Valence) Analysis: *Quantifying the positivity-negativity or emotions from a text data*

2 Topic Analysis or Modeling: *Detecting/classifying/clustering the main topics in a large textual data file.*

Sentiment Analysis



Topic Modeling



Text Analytics: Manual vs. Automated Tools

Practically Speaking: You can always do sentiment-topic analysis manually:

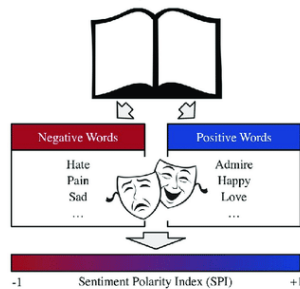
Simply by eye-check and giving your positivity scores for each tekst/review (sentiment) or coding which topics are there after reading the text

But most of the time **data-sets are too large** for that and it would take too much time + such assessments are subject to “human-error”

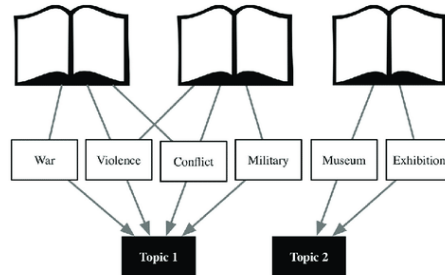
Reading and assigning scores for 50 online reviews is doable, 500 still could be OK if you have lot of time... but how about 5000, 50000 reviews?

Sentiment-Topic analysis **tools** do this for you in an **automated** way, in much shorter time by using **Machine Learning** based models/algorithms.

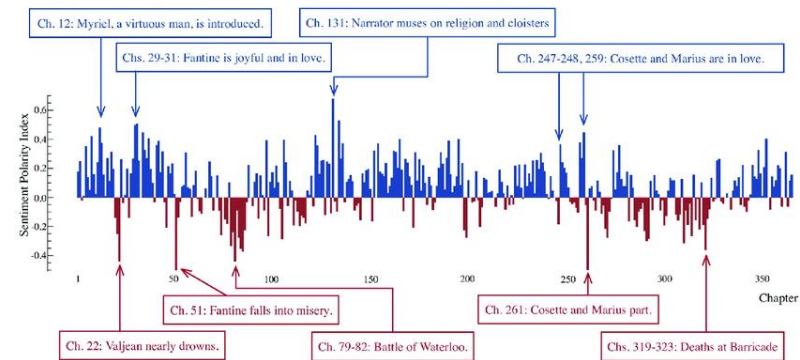
A Sentiment Analysis



B Topic Modeling



C Chapter Sentiments in Les Misérables



Text Analytics: Methods-Models & Tools

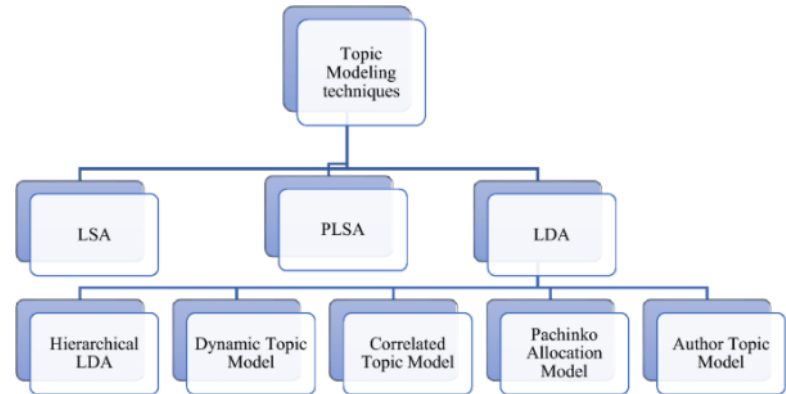
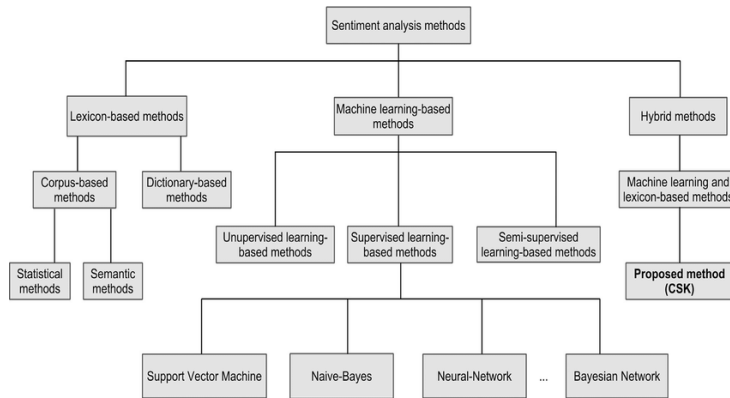
There are **many underlying methods-models** and techniques for sentiment-topic analysis.

These methods/techniques go **beyond the scope of this course**: a good background in statistics, econometrics and data science needed.

However to give an idea; here are what these methods are. Each has strengths & weaknesses:

Of course: you are not expected to know and apply all the advanced methods listed below

However you can learn the basics of sentiment-topic analysis from a practical perspective and apply/use them as Marketing Professionals !



Sentiment Analysis

Valence, Emotions, Polarity & Subjectivity

Sentiment Analysis: In a Nutshell

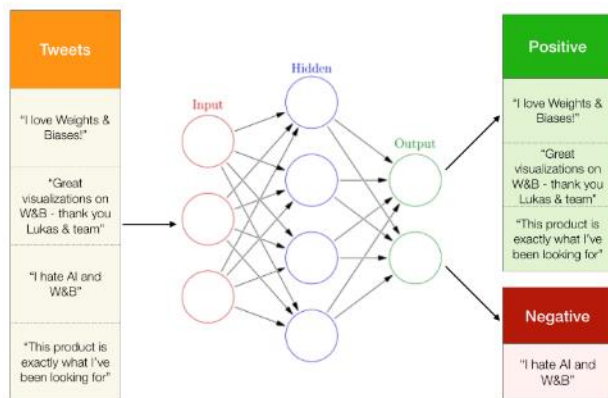
Sentiment Analysis (in practice) classifies the text data; reviews, social media posts etc. on the basis of their

(i) Positivity/Valence (ii) Polarity-Extremity (iii) Subjectivity/Emotionality or if more advanced (ii) Emotions – in detailed breakdown

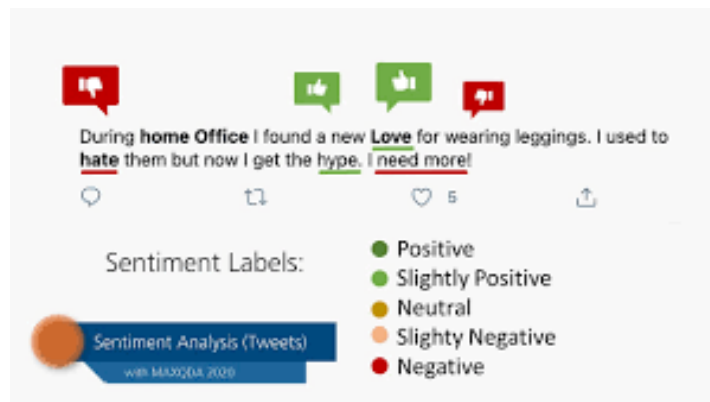
Sentiment Models/Techniques can be:

- 1 Key-Word (Lexicon) based: simple/easier, faster, more classification error (see the example on the right)
- 2 Contextual-Semantic based: more accurate – takes a look at the meaning in all sentence, not only the keywords (example: on left)

Context-Semantics (Machine Learning) Based



Keyword-Lexicon Based



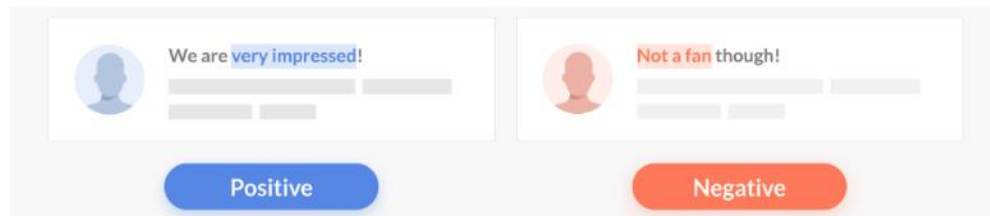
Sentiment Analysis: Positivity-Negativity or Valence

In most Sentiment Analysis tools-models, even with the simplest ones Positivity (or Valence) is the fundamental basis for classification

In this very fundamental step of analysis the text data is classified in different ways:

- 1 Positive-Neutral-Negative** simplistic – and very common in most free-tools
- 2 Valence/Positivity Scale** on a 4, 5, 7 or 10 point scale
- 3 Valence Score** gives more and accurate information such as: this review's positivity score is 83,2 out of 100 points

	Positive or Negative?
My coffee was great	positive
My coffee was awful	negative
My coffee was not great	negative
My coffee was not that great	neutral?
I did not think my coffee was great	negative
I did not expect my coffee to be this great	positive
I was disappointed with the quality of the coffee	negative
I was not disappointed with the quality of the coffee	positive



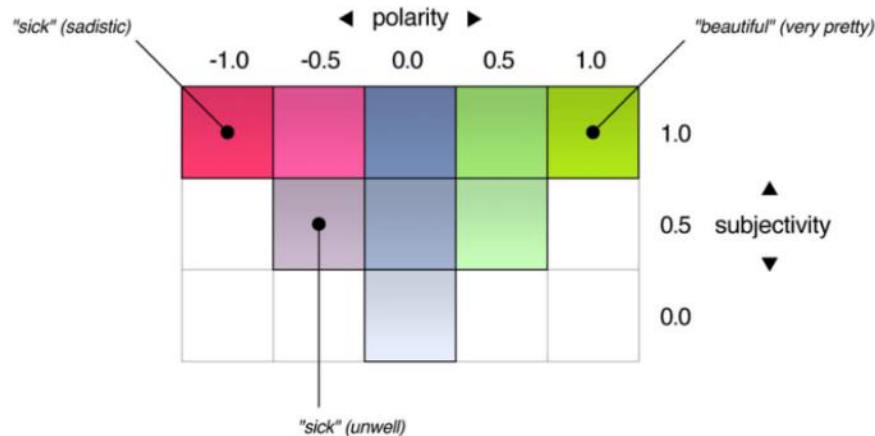
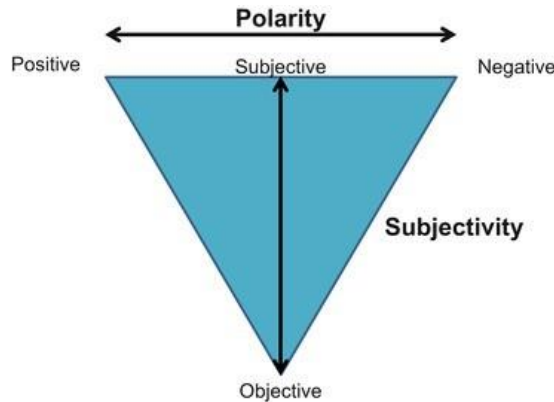
Sentiment Analysis: Subjectivity & Polarity

SUBJECTIVITY analysis classifies content into **objective (facts)** or **subjective (opinions)**

*In some NLP/sentiment models Subjectivity is also known-labeled as **Emotionality** where the Factual(ity) or Objectivity is on the opposite side of the scale*

POLARITY analysis indicates what is the **strength** / how **fierce** is of an opinion as being positive, neutral, or negative

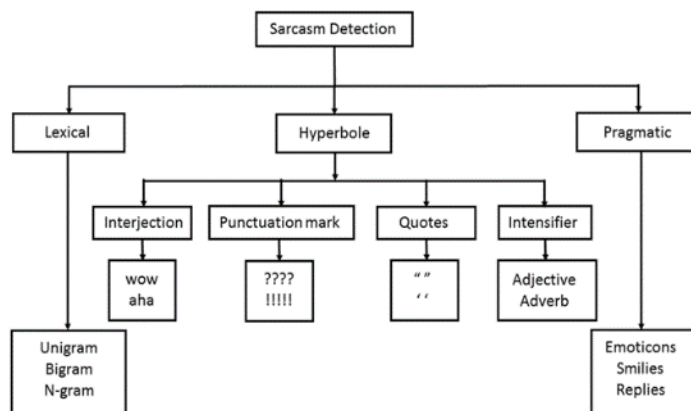
*In some NLP/sentiment models Polarity is also known-labeled as **Extremity**.*



Sentiment Analysis: Further Ahead > Sarcasm Detection

Sarcasm Detection, being a more subtle issue, can be elaborated with certain (i) clues (ii) by using different methods. See below:

Interjections, Punctuation marks, quotes etc. can be basis for sarcasm detection. By using Machine Learning methods more accurate models possible:



Sarcastic

Prediction

ye ##a i agree you are totally innocent and this is your first warning ever	0.928
this could never happen naturally , a clear indication of divine will !	0.915
well yeah ! cause you didn't ! even get into the war until it was basically already over !	0.901

Non-Sarcastic

Prediction

as a democrat i can say his first 100 days are a success .	0.185
one of the lost vikings	0.295
commented just to make it 322	0.384

Sentiment Analysis: Challenges – “Is it Really Neutral.. or What Does it Really Mean?”

Weakness of **Lexicon-Dictionary based** methods in Sentiment Analysis: they only look at **key-words** – mostly **not to the actual meaning or context**

For example: Take a look at the reviews below – particularly at the NEUTRAL (according to keyword based classification) one.

Do you think it is really: “NEUTRAL” – or does it sound more like POSITIVE to you ?

But still many firms use dictionary-keyword based methods: since they are easy, practical, less costly and requires less investment to data science

The diagram shows three sentiment categories arranged horizontally. Each category has a colored circular icon with a face, a label, and a sample review text.

- Negative:** A red circle with a sad face. The label "Negative" is below it. The review text is: "I'm dissatisfied with your customer service. No one was able to help me with the problems I had with using your product."
- Neutral:** A blue circle with a neutral face. The label "Neutral" is below it. The review text is: "The product has multiple features that are suitable for users with different levels of experience." This category is enclosed in a dashed blue rounded rectangle. A blue arrow points from the bottom right of this rectangle to a blue box containing the text "Do you think this is really neutral?".
- Positive:** A yellow circle with a happy face. The label "Positive" is below it. The review text is: "I really enjoy how easy this product is to use and how it successfully helps my team complete their day-to-day tasks."

Topic Modelling

Which Topics Are Prominent & Have Priority

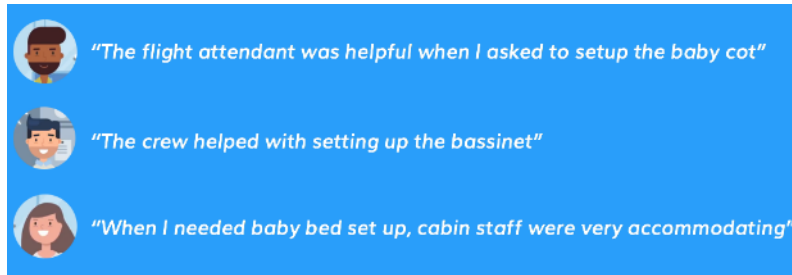
Topic Models: Classifying-Clustering Topics

Topic Modeling (Analysis) is mostly classifying text-data on the basis of main topics covered/mentioned.

Imagine you have thousands of customer reviews or transcribed customer service calls... which is impossible to read/interpret by human force

As a marketing manager you may wish to check the prominence of the topics by checking customer product/service reviews and make decisions on

- (i) Which topics should be priority for you – Which main topics your customers are mainly talking about
- (ii) How can you approach them / What could be the solutions?



Three customer reviews are displayed, each with a circular avatar of a person. The reviews are:

- "The flight attendant was helpful when I asked to setup the baby cot"
- "The crew helped with setting up the bassinet"
- "When I needed baby bed set up, cabin staff were very accommodating"

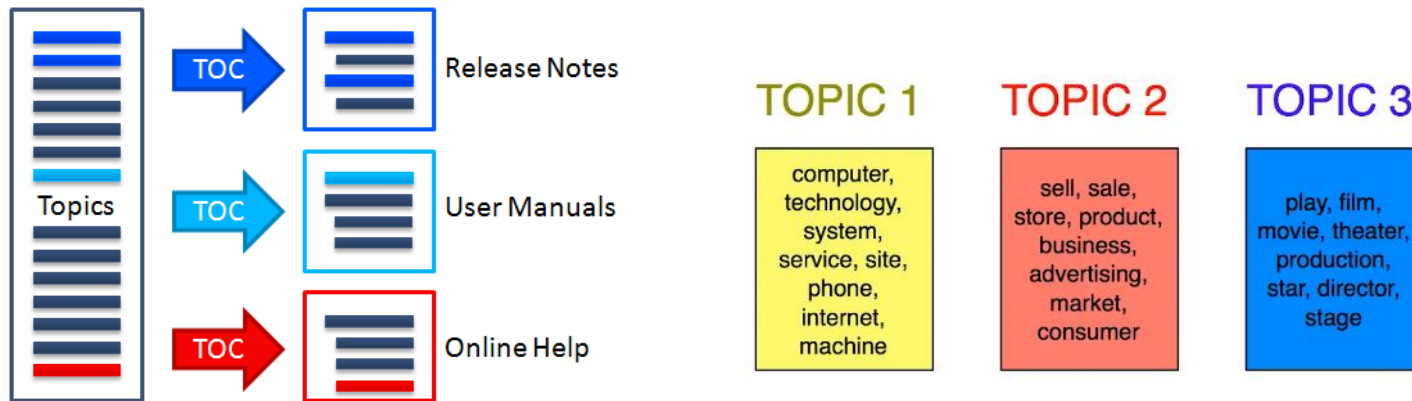


Topic Models: Classifying-Clustering Topics

Like most text-analytics work the simplest way is to read and classify them manually (by reading/interpreting):

But what if you have a very large data set – or tens or hundreds of various topics mentioned?

- | | |
|-------------------------------------|---|
| 1 Manual / Keyword Based Analysis: | simple, accurate, less efficient-problematic with large data |
| 2 Latent Dirichlet Allocation (LDA) | sophisticated, maybe less accurate, good with large data sets |
| 3 Structural Topic Model (STM) | sophisticated, more accurate, good with large data sets |

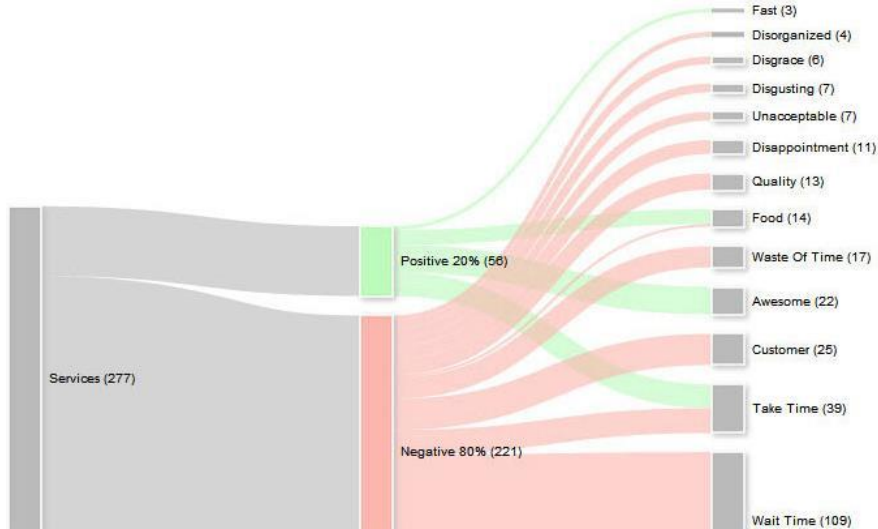


Sentiment & Topic Analysis: Use in Combination

In this example through Social Media posts (i) Sentiment and (ii) Topic Analysis in combination to investigate:

Customers' sentiment positivity-negativity on different topics: What are the main TOPICS that customers are HAPPY/ NOT SO HAPPY with ?

Think: How useful this could be in Customer Service Management



Sentiment & Topic Analysis: Use in Combination

Shall we try it? Let's have a practical and non technical example:

List yours' sentiment positivity-negativity on different topics: What are the main TOPICS related to NETFLIX that you are positive/negative LIKE/DISLIKE ?

(List 2-3 topics per sentiment)

NETFLIX

Sentiment & Topic/text Analysis: Tools

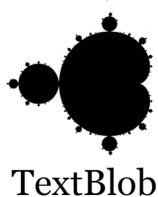
Many commercial and academic tools, software, models and dashboards are available for sentiment analysis (you can choose among many options)

However we take a quick look at some tools that you can use for academic purposes:

1. Evaluative Lexicon



2 Python TextBlob



3 Hugging Face – Zero-shot



Hugging Face

FOR THIS
COURSE

4 Python NLTK/ Spacy



5 R Tidytext/ STM/ LDA



OPTIONAL
FOR FURTHER
LEARNING

Sentiment & Topic Analysis Practice

With Evaluative Lexicon

Evaluative Lexicon *etc. Based, Scientifically Validated NLP Software*

Evaluative Lexicon (EL) is a lexicon-based scientifically validated text-analytics and processing software. Free for academics and students.

EL classifies **valence, extremity, and emotionality** of individuals' opinions in language.

*To construct Evaluative Lexicon dictionary, researchers used **billions of words, millions of online reviews, and the judgments of a large set of external raters.***

*Based on this data-driven approach, the EL searches only for words that provide a **reliable signal of individuals' opinions in natural language.***



More advanced lexicon based sentiment detection with higher level of quantification (valence, extremity, emotionality scores) Academically validated. Has its limitations (lexicon based)

Open Files for Analysis...

Open Folder of Files...

Behav Res
DOI 10.3758/s13428-017-0975-6



The Evaluative Lexicon 2.0: The measurement of emotionality, extremity, and valence in language

Matthew D. Rocklage¹ · Derek D. Rucker¹ · Loran F. Nordgren¹

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Abstract The rapid expansion of the Internet and the availability of vast repositories of natural text provide researchers with the immense opportunity to study human reactions, opinions, and behavior on a massive scale. To help researchers take advantage of this new frontier, the present work introduces and validates the Evaluative Lexicon 2.0 (EL 2.0)—a quantitative linguistic tool that specializes in the measurement of the emotionality of individuals' evaluations in text. Specifically, the EL 2.0 utilizes natural language to measure the emotionality, extremity, and valence of evaluative reac-

wordlist and normative emotionality, extremity, and valence ratings are freely available from www.evaluativelexicon.com.

Keywords Emotion · Text analysis · Language · Attitudes · Cognition

Over the last decade, the rapid expansion of the Internet and the availability of vast repositories of natural text have allowed researchers to systematically

Evaluative Lexicon *Lexicon Based, Scientifically Validated NLP Software*

The use of EL is quite easy and practical;

Demo: After installing-activating the software I simply upload the demo batch of reviews (about Amsterdam Hilton) in the software;

NOTE: It would be wise to remove the “commas” in your tekst data before you load your file as EL software can recognize commas as separators.

ID	Review
1	The room was spacious, had a nice view, the food was excellent, just as I remembered when I also stayed in the hotel a few years ago.
2	The bathroom was okay, clean, and everything I needed was there, except, I think they should modernize and make the bathroom
3	Amazing view on the canal, the breakfast was good, rooms are super clean.Helpful and kind staff, very good location.
4	The italian restaurant is overpriced and not that good. There was a very strange, unpleasant smell in the corridors, like burning p
5	Super hotel. No weak points.
6	The sun terrace was stunning! And the food was excellent! Location was perfect close to museums and cafes. Public transport w
7	Service is brilliant. Staff is excellent, helpful and cheerful. Thank you for the perfect stay.
8	Design is already a bit oldfashioned. Strong security measures on the elevator make it impossible to move from one floor to ano
9	The large rooms and the marvelous view over the city. The nice garden along the canal.
10	The initial check in was with a trainee. Although I appreciate everyone needs to learn - she wasn't trained in smiling or welcomin
11	Super supportive in terms of early check-in / late check-out which helped our already very disrupted travel schedule at the time.
12	The prices for food and drinks on their very nice terrace are pretty punchy but likely in line with a 5-star hotel as well as this bran
13	The room was worth the price with really comfortable beds. The service was as expected, really good. Me and my brother was c
14	Service at the terrace was a bit slow when full bathroom ceiling had some stains from moisture, but nothing too serious
15	Only problem during the stay was when a busload of old people checked in and the number of lifts (elevators) was completely in

File Analysis Wordlists Help

ANALYSIS CSV/EXCEL PREVIEW SETTINGS

ID	Review
1.0	The room was spacious, had a nice view, the food was excellent, just as I remembered when ...
2.0	The bathroom was okay, clean, and everything I needed was there, except, I think they shoul...
3.0	Amazing view on the canal, the breakfast was good, rooms are super clean.Helpful and kind ...
4.0	The italian restaurant is overpriced and not that good. There was a very strange, unpleasant ...
5.0	Super hotel. No weak points.
6.0	The sun terrace was stunning! And the food was excellent! Location was perfect close to mus...
7.0	Service is brilliant. Staff is excellent, helpful and cheerful. Thank you for the perfect stay.
8.0	Design is already a bit oldfashioned. Strong security measures on the elevator make it impos...
9.0	The large rooms and the marvelous view over the city. The nice garden along the canal.
10.0	The initial check in was with a trainee. Although I appreciate everyone needs to learn - she w...
11.0	Super supportive in terms of early check-in / late check-out which helped our already very di...
12.0	The prices for food and drinks on their very nice terrace are pretty punchy but likely in line w...
13.0	The room was worth the price with really comfortable beds. The service was as expected, rea...
14.0	Service at the terrace was a bit slow when full bathroom ceiling had some stains from moistu...
15.0	Only problem during the stay was when a busload of old people checked in and the number...

Please select the column where the text is that you wish to analyze, then click 'Run Analysis'. [Preview look wrong?](#)

Run Analysis

After uploading the excel file with “hotel reviews” (text data)

- 1) I simply choose which is the column with text data to be analyzed
- 2) Then I click “Run the Analysis” button

Evaluative Lexicon *Lexicon Based, Scientifically Validated NLP Software*

After you run the analysis EL software will produce the processed (analyzed-results) xls. sheet to you as a different file.

There you will see each hotel review (customer comment) text is automatically coded and quantifies on the basis of following dimensions:

	A	B	C	D	E	F	G	H
1	ORIGINAL TEXT	ID	wordcount	valence_min	valence_max	valence_avg	valence_std	valence_min_pos
2	The room was spacious, had a nice view, the food was excellent	1	27	7.76	8.16	7.96	0.283	7.76
3	The bathroom was okay, clean, and everything I needed	2	39					
4	Amazing view on the canal, the breakfast was good, room	3	19	7.57	7.97	7.783	0.201	7.57
5	The Italian restaurant is overpriced and not that good. The	4	22	1.47	1.6	1.535	0.092	
6	Super hotel. No weak points.	5	5	7.57	7.57	7.57		7.57

VALENCE
How positive-negative each review is

	A	P	Q	R	S	T	U
1	ORIGINAL TEXT	extremity_min	extremity_max	extremity_avg	extremity_std	extremity_min_pos	extremity_max_pos
2	The room was spacious, had a nice view, the food was excellent	3.26	3.66	3.46	0.283	3.26	
3	The bathroom was okay, clean, and everything I needed						
4	Amazing view on the canal, the breakfast was good, room	3.07	3.47	3.283	0.201	3.07	
5	The Italian restaurant is overpriced and not that good. The	2.9	3.03	2.965	0.092		
6	Super hotel. No weak points.	3.07	3.07	3.07		3.07	
7	The sun terrace was stunning! And the food was excellent	2.96	3.71	3.498	0.359	2.96	3.71

EXTREMITY (POLARITY)
How fierce/intense the statements are

	A	AC	AD	AE	AF	AG
1	ORIGINAL TEXT	emotionality_min	emotionality_max	emotionality_avg	emotionality_std	emotionality
2	The room was spacious, had a nice view, the food was excellent		4.44	4.61	4.525	0.12
3	The bathroom was okay, clean, and everything I needed					
4	Amazing view on the canal, the breakfast was good, room		3.43	6.6	4.837	
5	The Italian restaurant is overpriced and not that good. The		2.7	5.21	3.955	
6	Super hotel. No weak points.		4.48	4.48	4.48	
7	The sun terrace was stunning! And the food was excellent		3.91	5.76	4.638	
8	Service is brilliant. Staff is excellent, helpful and cheerful		3.83	6.85	4.576	
9	Design is already a bit oldfashioned. Strong security measures		3.46	3.46	3.46	
10	The large rooms and the marvelous view over the city. The		4.61	6.14	5.375	1.082
11	The initial check in was with a trainee. Although I appreciated		4.92	5.62	5.323	0.362
12	Super supportive in terms of early check-in / late check-out		4.48	5.24	4.82	0.386

EMOTIONALITY (SUBJECTIVITY)
How subjective (emotional) vs factual each review is

Evaluative Lexicon *Lexicon Based, Scientifically Validated NLP Software*

How to use

The software can utilize TXT, CSV, and Excel files and can analyze millions of full-text records.

We hope that the program itself should be intuitive to use, but we've created a document that guides you through each variable the LS calculates ([link](#)).

See our academic publications ([Evaluative Lexicon](#), [Certainty Lexicon](#)) which provide more detail on how we constructed and validated the LS tools as well as pointers on how best to utilize the measured variables.

The LS was specifically designed to be a general text analysis tool - one that focuses on a person's opinion and on language that is used consistently across a large range of topics. In that regard, the LS emphasizes accuracy over absolute coverage and thus will not code a piece of text if it does not contain an LS word. See this chapter we wrote as an example of the tradeoffs of the different text analysis approaches ([link](#), pp. 18-20).

We have tested the software extensively, but it is still in beta so please let us know if you run into any issues by e-mailing Matt Rocklage at m.rocklage@northeastern.edu.

Via the website, click here to read the variables meaning



Evaluative Lexicon (EL) output:

- **wordcount:** number of words in that text
- **Valence**
 - o **valence_min:** the value of the least positive (most negative) word in the text
 - o **valence_max:** the value of the most positive word in the text
 - o **valence_avg:** the weighted average valence of the text
 - Note: the average is weighted for those EL words used more than once in the text (see Rocklage, Rucker, and Nordgren 2018 for more)
 - o **valence_std:** if more than one EL word is used, the weighted standard deviation
 - o **valence_min_pos:** the least positive word that is above the midpoint of the EL scale (i.e., above 4.50 out of 9.00)
 - o **valence_max_pos:** the most positive word that is above the midpoint of the EL scale
 - o **valence_avg_pos:** the weighted average valence of those words above the midpoint of the EL scale

Text Analytics – Python

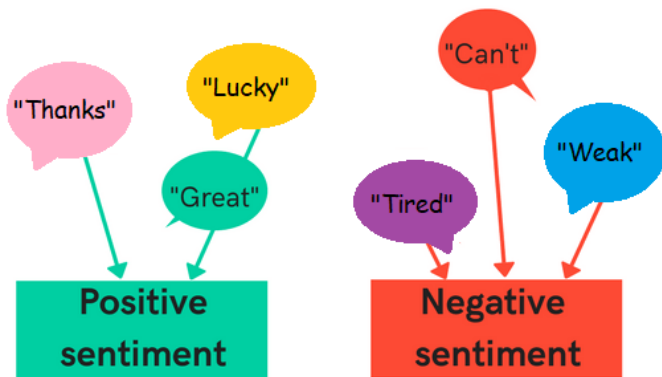
Sentiment Analysis with Text Blob

Topic or Emotion Classification with Hugging Face (Zero Shot Classifier)

Tekst Analytics with Python (Libraries: TextBlob, NLTK, Spacy, Gensim)

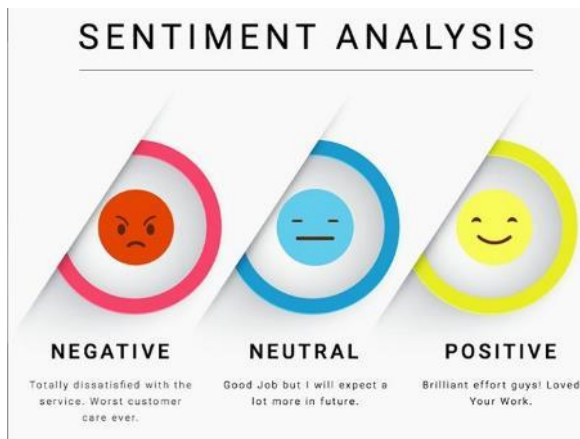
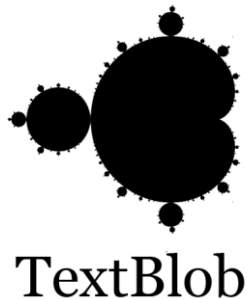
Python has also very strong libraries for Natural Language Processing, text analytics and in particular sentiment analysis

I would like to show you a simple example with Text Blob – to demonstrate it could be easier than you think (so you can use in future)



Python: TextBlob Library – General Information

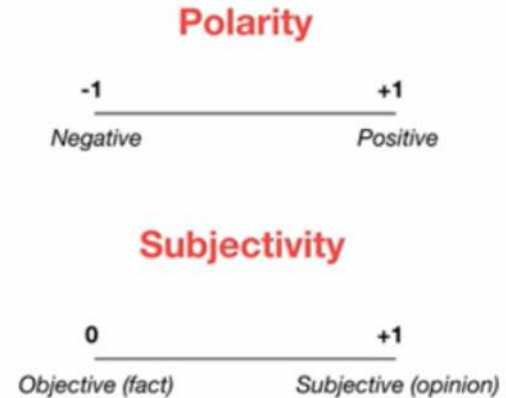
The TextBlob library is a popular Python library for natural language processing (NLP). It provides a simple and easy-to-use interface for performing a variety of NLP tasks such as as part-of-speech tagging, noun phrase extraction, sentiment analysis, classification, and translation.



- In digital marketing, the TextBlob library can be used in a number of ways to analyze customer feedback, social media posts, and other types of text data.
- For example, digital marketers use TextBlob to perform sentiment analysis on customer reviews:
- By analyzing the sentiment of customer reviews they gain insights into how customers feel about their products. This can help them identify areas for improvement and make data-driven decisions to improve customer satisfaction.

Source: <https://textblob.readthedocs.io/en/dev/>

Python: TextBlob Library – General Information



In TextBlob library, the **polarity score ranges from -1 to +1**, where -1 means very negative sentiment, 0 means neutral sentiment, and +1 means very positive sentiment. Similarly, **subjectivity score also ranges from 0 to 1**, where 0 means the text is very objective or factual, and 1 means the text is very subjective or opinionated.

Sentiment with Python: A Simple Example with TextBlob

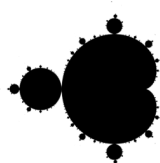
By using Python libraries such as TextBlob you can do many things with text data:

- 1 Pre-Processing Text Data: splitting the sentence into words, large text into sentences, defining words as noun, verb etc.
- 2 Preparing for Analysis: stemming/lemmatizing, eliminating (deleting) the stop-words, short words, or private nouns
- 3 Sentiment Analysis: analyzing the positivity (+/0) of the text but also subjectivity and polarity
- 4 Topic Modeling/Analysis: classifying text (social media posts or comments) based on their topic

Command Prompt

```
Microsoft Windows [Version 10.0.18363.1801]
(c) 2019 Microsoft Corporation. All rights reserved.

H:\>pip install textblob_
```



TextBlob

```
Python 3.9.5 (tags/v3.9.5:0a7dcbd, May 3 2021, 17:27:52) [MSC v.1928 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> from textblob import TextBlob
>>> hotel = TextBlob("This hotel is great. The location was perfect and the breakfast was delicious")
>>>
```

Like usual you start by PIP installing TextBlob library from your command prompt

Now I import my text data: Here it is a customer review about an hotel.
NOTE: You don't have to enter a text. You can also open/import an excel or csv file where many comments-posts are stored.

Sentiment with Python: A Simple Example with TextBlob

If we find the standard analyser too simple – we can change/use different Sentiment Analyzers built in TextBlob.

Here I import and use Naïve Bayes Analyzer. This gives me more accurate-results on sentiment:

```
>>> from textblob import TextBlob
>>> from textblob.sentiments import NaiveBayesAnalyzer
>>> blob = TextBlob("This hotel is great. The location was perfect and the breakfast was delicious", analyzer=NaiveBayesAnalyzer())
>>> blob.sentiment
Sentiment(classification='pos', p_pos=0.8498264033211297, p_neg=0.15017359667886998)
>>> _
```

	User_ID	Description	Browser_Used	Device_Used	Is_Response
0	id10326	The room was kind of clean but had a VERY stro...	Edge	Mobile	not happy
1	id10327	I stayed at the Crown Plaza April -- - April -...	Internet Explorer	Mobile	not happy
2	id10328	I booked this hotel through Hotwire at the low...	Mozilla	Tablet	not happy
3	id10329	Stayed here with husband and sons on the way t...	InternetExplorer	Desktop	happy
4	id10330	My girlfriends and I stayed here to celebrate ...	Edge	Tablet	not happy

OR you can even classify Hotel Reviews as being happy/not happy
By using a classifier (within TextBlob)

Sentiment with Python: A Simple Example with TextBlob

```
>>> from textblob import TextBlob
>>> hotel = TextBlob("This hotel is great. The location was perfect and the breakfast was delicious")
>>> hotel.tags
[('This', 'DT'), ('hotel', 'NN'), ('is', 'VBZ'), ('great', 'JJ'), ('The', 'DT'), ('location', 'NN'), ('perfect', 'JJ'), ('and', 'CC'), ('the', 'DT'), ('breakfast', 'NN'), ('was', 'VBD'), ('delicious', 'JJ')]
>>> _
```

Here as a simple example: I start by tagging the words one-by-one in review with “tags” command
You can see that each word is separately tagged and word type identified.

```
>>> hotel.sentiment
Sentiment(polarity=0.9333333333333332, subjectivity=0.9166666666666666)
>>> _
```

SENTIMENT ANALYSIS

Next with only one command (adding: .sentiment)
I get the sentiment analysis results of this review
Polarity: 0,933 positive-very high
Subjectivity: 0,916 – thus very subjective

```
>>> hotel.words
WordList(['This', 'hotel', 'is', 'great', 'The', 'location', 'was', 'perfect', 'and', 'the', 'breakfast', 'was', 'delicious'])
```

```
>>> hotel.sentences
[Sentence("This hotel is great."), Sentence("The location was perfect and the breakfast was delicious")]
```

FINALLY to demonstrate how practical-fast this is
I now split the
Sentences into words (adding .words)
Review into sentences (adding: .sentences)
By using many TextBlob command like this you can play with/analyze your text data

Sentiment Analysis with TextBlob

Real Data Practice on Google Colab

Further Data Analysis TextBlob Library – Shall we try it?



 DMA Tutorial2_sentiment.ipynb ☆

File Edit View Insert Runtime Tools Help [Last edited on 5 April](#)

+ Code + Text

Tutorial 2 - Sentiment Analysis

This code uses the TextBlob library to perform sentiment analysis on a CSV file containing text data of brands. Sentiment analysis is a valuable tool for digital marketers as it helps them understand how the audience perceives the brand, which sentiment the brand uses to communicate, and ultimately, how they can improve their digital marketing strategies.

PS.: Before running the cells, upload the data. You can upload data by following these steps:

1. Click on the "Files" tab on the left-hand side of the screen.
2. Click on the "Upload" button.
3. Select the file you want to upload from your computer.

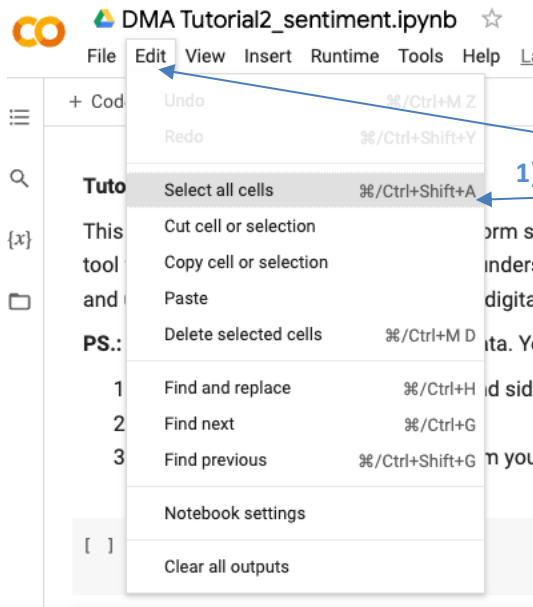
```
[ ] !pip install pandas
    !pip install textblob
```

The TextBlob library is used to calculate the subjectivity and polarity of each text. Subjectivity is how subjective or objective the text is, while polarity is the sentiment expressed in the text, whether it is positive, negative, or neutral

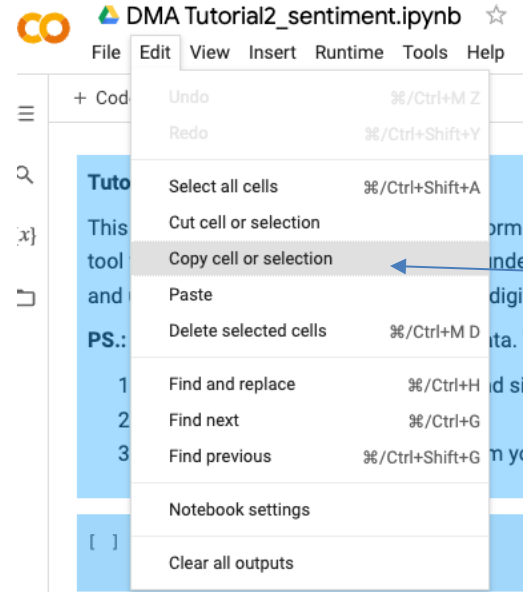
[Click here](#)



TextBlob Library – Step 1 - Making your own copy of the notebook

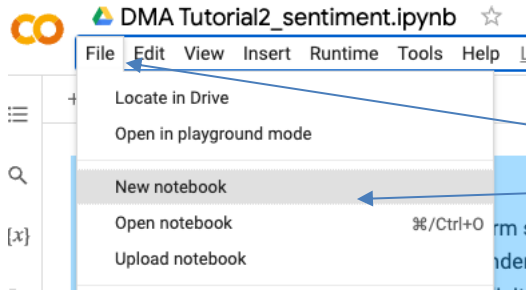


1) Click in EDIT > Select all cells

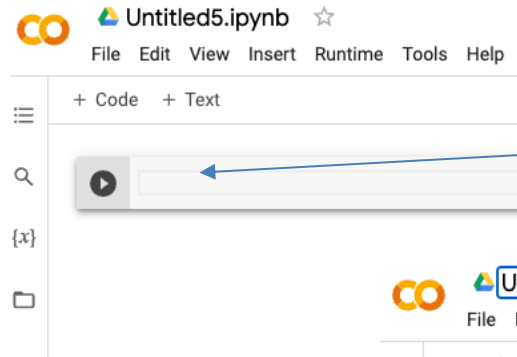


2) Click in copy cell or selection

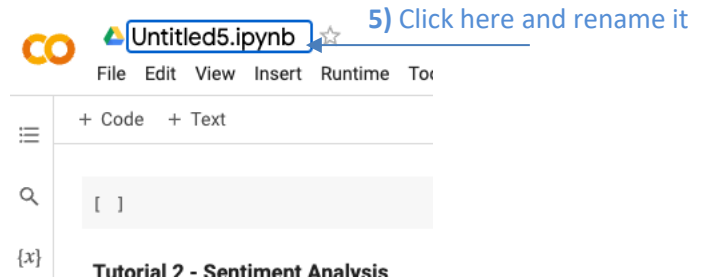
TextBlob Library – Step 1 - Making your own copy of the notebook



3) Click in File > New notebook



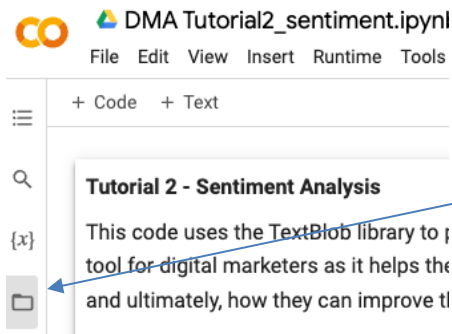
4) Paste in the new cell the content



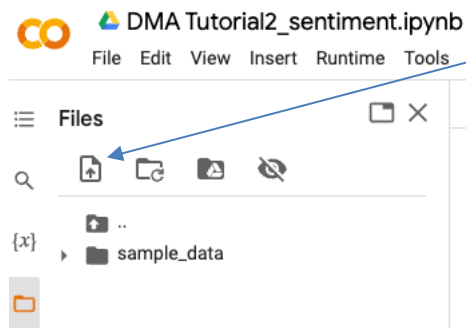
5) Click here and rename it

TextBlob library – Step 2 - Let's upload the dataset

1) Download the sentiment dataset from CANVAS TO YOUR COMPUTER

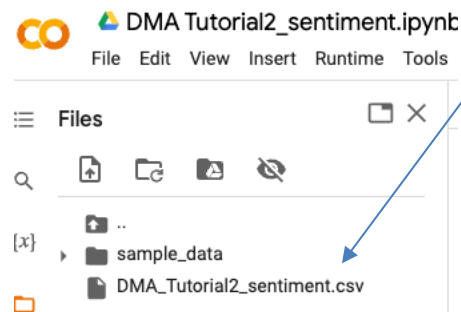


2) Click in the File folder icon



3) Click here to upload your dataset

4) Well done! Your data is here



TextBlob library – Step 3 – Press the play

Tutorial 2 - Sentiment Analysis

This code uses the TextBlob library to perform sentiment analysis, a tool for digital marketers as it helps them understand how customers feel about their brand and ultimately, how they can improve their digital marketing strategy.

PS.: Before running the cells, upload the data. You can upload data in two ways:

1. Click on the "Files" tab on the left-hand side of the notebook.
2. Click on the "Upload" button.
3. Select the file you want to upload from your computer.

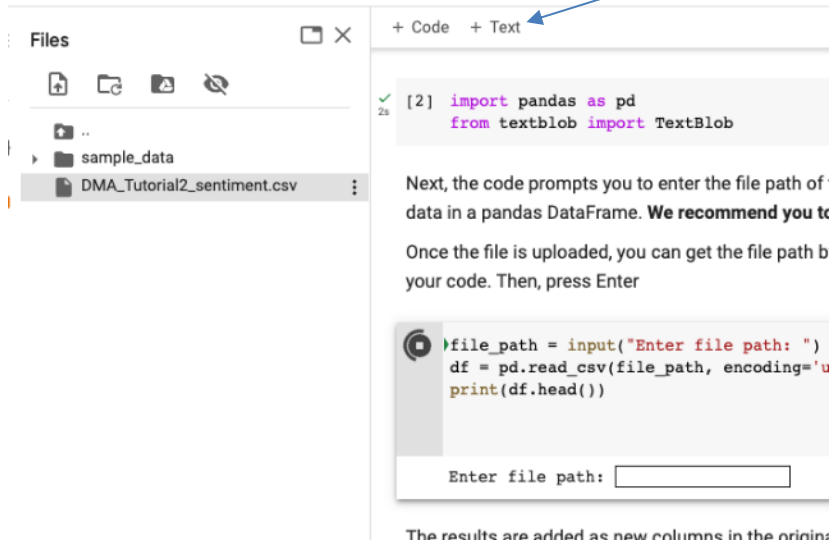
 `!pip install pandas`

Run cell (⌘/Ctrl+Enter)
cell has not been executed in this session

The TextBlob library is used to calculate the subjectivity and polarity of the sentiment expressed in the text, whether it is positive, negative, or neutral.

1) Press here and the code in this cell will run/start

Some cells contain text and other code. We will run the ones with a code



The screenshot shows a Jupyter Notebook interface. On the left, the 'Files' tab is active, displaying a file named 'DMA_Tutorial2_sentiment.csv' under the 'sample_data' directory. The main area shows a code cell with the following code:

```
[2] import pandas as pd
from textblob import TextBlob
```

Below the code cell, there is a text input field for the file path, followed by a code cell with the following code:

```
file_path = input("Enter file path: ")
df = pd.read_csv(file_path, encoding='utf-8')
print(df.head())
```

The input field contains the text 'Enter file path: '.

The results are added as new columns in the original DataFrame.

TextBlob library – Step 4 – How to get the file path?

The notebook will ask you to enter the file path. File path is the location of your data.

1) Click the 3 dots next to your data

Files

- ..
- sample_data
 - DMA_Tutorial2_sentiment.csv

```
[2] import pandas as pd
from textblob import TextBlob
```

Next, the code prompts you to enter the file path of data in a pandas DataFrame. **We recommend you to** Once the file is uploaded, you can get the file path by your code. Then, press Enter

```
file_path = input("Enter file path: ")
df = pd.read_csv(file_path, encoding='u
print(df.head())
```

Enter file path:

The results are added as new columns in the origin:

Files

- ..
- sample_data
 - DMA_Tutorial2_sentiment.csv

- Download
- Rename file
- Delete file
- Copy path
- Refresh

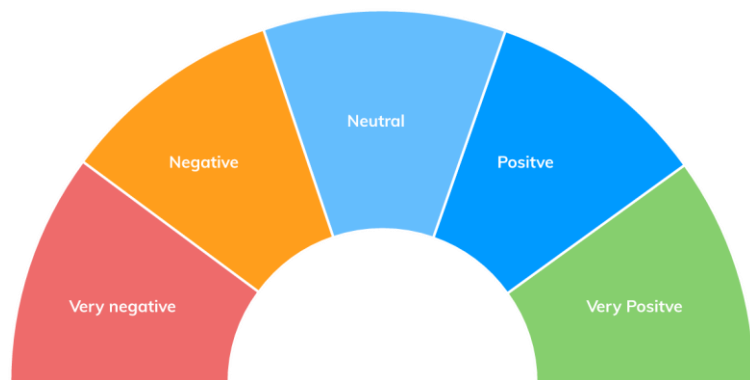
2) Click Copy path

3) Paste here the file path

TextBlob library – sentiment analysis

Polarity is the output that lies between $[-1,1]$, where -1 refers to negative sentiment and +1 refers to positive sentiment. Subjectivity is the output that lies within $[0,1]$ and refers to personal opinions and judgments.

How would you describe your initial impressions about the brand's communication on Twitter based on the results of sentiment analysis using TextBlob?



Text (Topic & Emotion) Classification

w/ Hugging-Face (Zero-Shot Classifier)

Zero-Shot Classification: Classifying Tekst Data

Zero-shot text classification is a natural language processing technique enabling machine learning model to classify text into pre-defined categories without being trained on those categories.

The model can make accurate predictions about the meaning or topic of text even if it has never seen examples of that category before.



Hugging Face

Hugging Face is a popular natural language processing (NLP) company that provides open-source libraries and pre-trained models for a variety of NLP tasks.



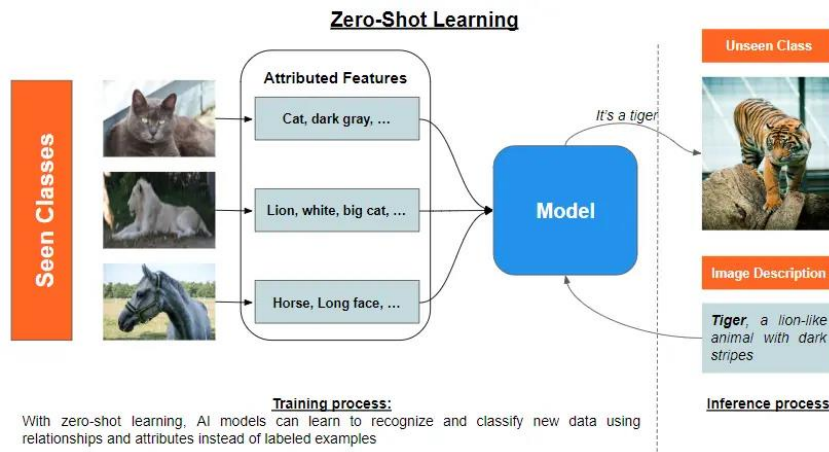
Zero-Shot Classification: Classifying Tekst Data

Zero-shot text classification can be a valuable tool for digital marketers to analyze and classify large amounts of text data, such as customer feedback, reviews, and social media posts.



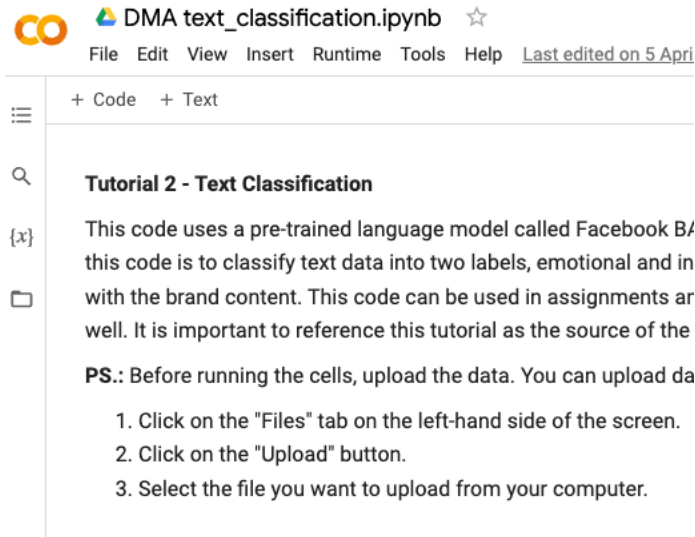
Hugging Face

Digital marketers can use zero-shot text classification to classify text data into different categories, such as product features, service issues, or pricing. This can help them better understand what customers are talking about and identify areas for improvement.



By analyzing the results of the classification, digital marketers can gain insights into which types of brand language and appeals are most effective in eliciting positive reactions from customers. They can then use this information to develop effective marketing campaigns,, and brand positioning.

Zero-Shot Text Classification – Shall we start it?



The screenshot shows a Jupyter Notebook titled "DMA text_classification.ipynb". The interface includes a top menu bar with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu bar, there are tabs for "+ Code" and "+ Text". The notebook content is titled "Tutorial 2 - Text Classification" and contains the following text:

This code uses a pre-trained language model called Facebook B/ this code is to classify text data into two labels, emotional and in with the brand content. This code can be used in assignments ar well. It is important to reference this tutorial as the source of the

PS.: Before running the cells, upload the data. You can upload da

1. Click on the "Files" tab on the left-hand side of the screen.
2. Click on the "Upload" button.
3. Select the file you want to upload from your computer.



Hugging Face

[1\) Click here](#)

[2\) Follow the same steps as we did for the sentiment analysis notebook](#)

Zero-shot text classification – Shall we try something new?



Hugging Face

The text in each row of the DataFrame is then classified using the zero-shot classifier, which returns the score for each label (emotional and informational) for each row. The code stores the scores and best label for each row in separate lists.

```
[ ] # classify the text using the zero-shot classifier
    labels = ["emotional", "informational"]
    scores = []
    best_labels = []
```



Which other types of labels, in addition to the informational versus emotional labels, do you think would be beneficial to apply when performing text classification using zero-shot on your company's data?

Appendix (Optional)

Text Analytics with R Libraries

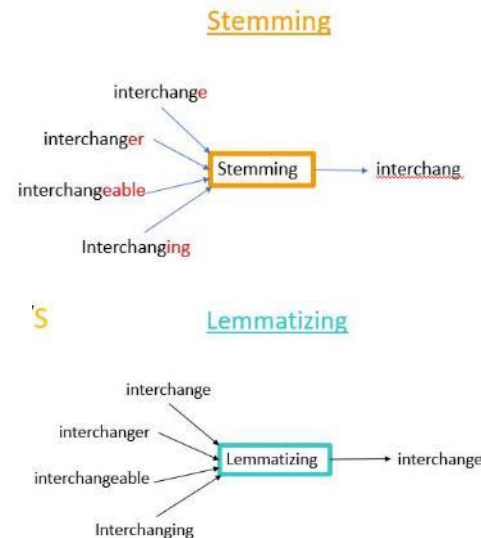
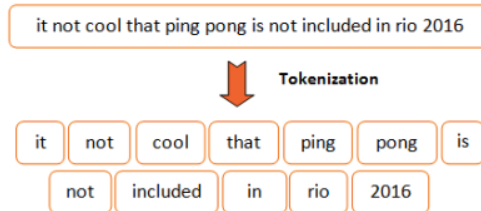
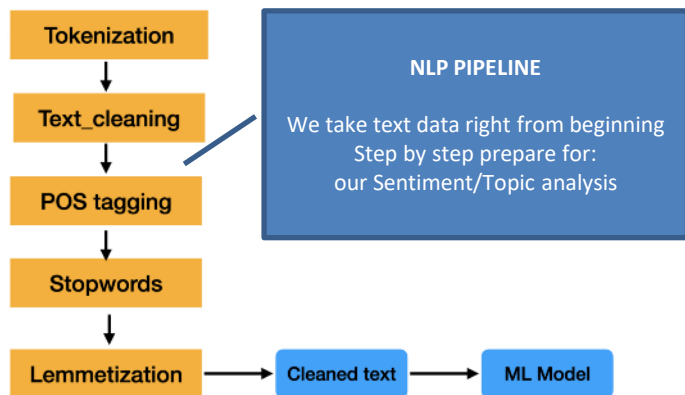
*Provides some examples and references for R Based Text Analytics
No practical application is expected.*

Text Analytics with R/ Python: Natural Language Processing

Not required in this course but if you are interested in advanced text analytics;

Using R or Python (related libraries) for text analytics enable you to use Natural Language Processing Methods and take the full control and do many things in advanced text analytics:

NLP Pipeline:

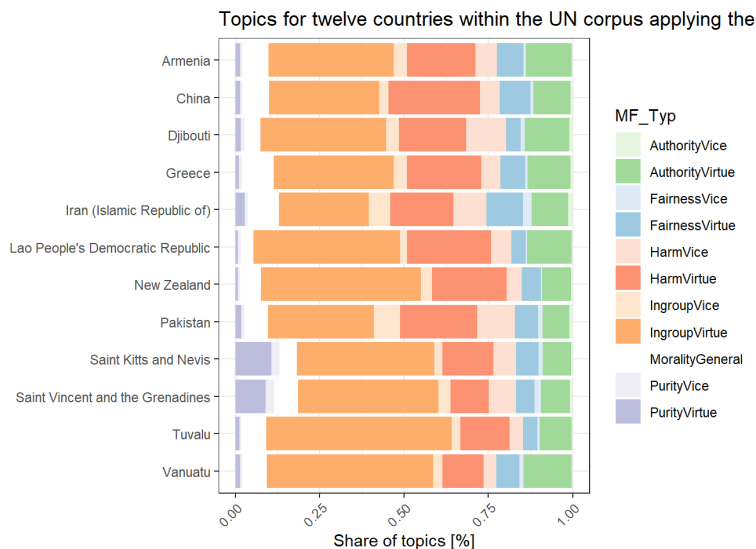
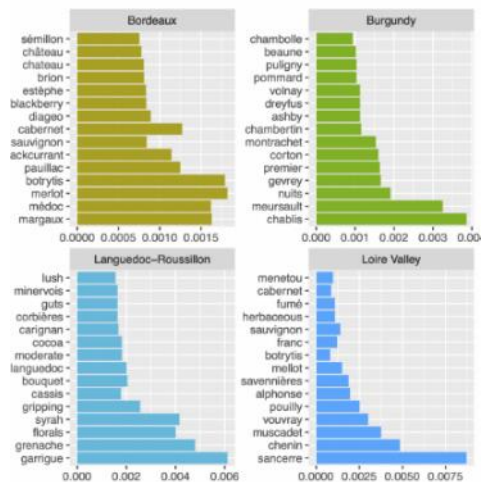


Sentiment/Topic Analysis Tools – R (Libraries: tidytext, Quanteda, SpacyR)

R packages such as tidytext, **Quanteda**, **SpacyR** will allow you to do a lot more with very large text datasets by using NLP.

Check the example below: How different French wine regions are characterized with certain words – as a classification (150.000 reviews!)

But keep in mind: Learning curve for R based SA is steep and will be time consuming (will pay-off if you are patient/persistent though)



quanteda
Quantitative Analysis of Textual Data

Sentiment/Topic Analysis Tools – R (Libraries: tidytext, Quanteda, SpacyR)

Apart from analytical libraries there are also **lexicons-dictionaries** we use in text-analytics in R:

Here three most commonly used R sentiment lexicons (dictionaries) for words, word groups and their sentiments:

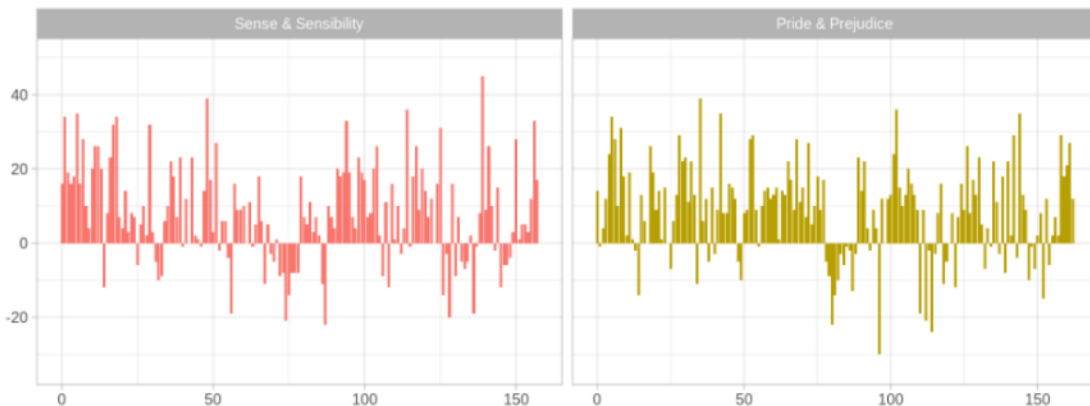
- **AFINN** from Finn Årup Nielsen,
- **bing** from Bing Liu and collaborators, and
- **nrc** from Saif Mohammad and Peter Turney.

```
library(tidytext)

get_sentiments("afinn")
```

For sentiment analysis you can use any of them (as you wish) and self-assess their performance: example – sentiment in Jane Austen novels

##	word	value
##	<chr>	<dbl>
##	1 abandon	-2
##	2 abandoned	-2
##	3 abandons	-2
##	4 abducted	-2
##	5 abduction	-2
##	6 abductions	-2
##	7 abhor	-3
##	8 abhorred	-3
##	9 abhorrent	-3
##	10 abhors	-3

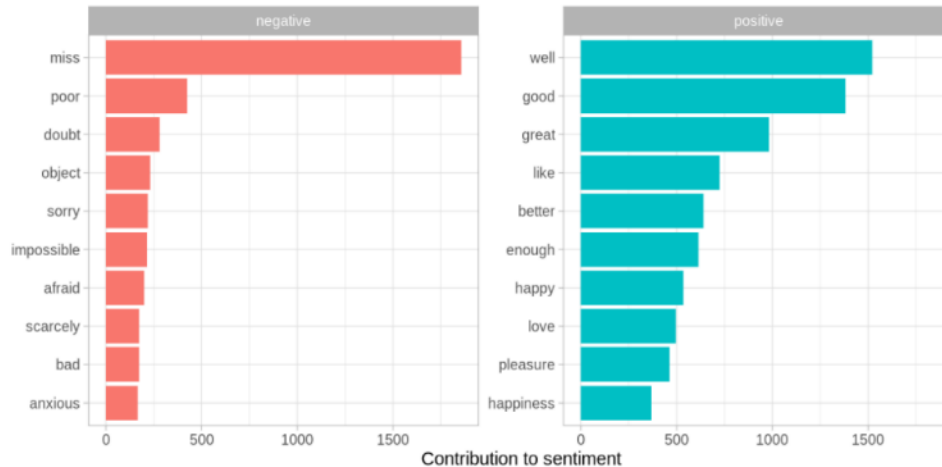


Sentiment/Topic Analysis Tools – R (Libraries: tidytext, Quanteda, SpacyR)

Here are some examples of what you can do with R (tidytext) combined with lexicons:

- 1 Most common positive and negative words in Jane Austen novels – based on “bing” lexicon
- 2 A word-cloud based on most common words in Jane Austen novels

If you want to practice codes/scripts are available on: <https://www.tidytextmining.com/sentiment.html>



Thank You